

Three Steps to More Efficient 3G, Base Station Power Amplifiers

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Outline

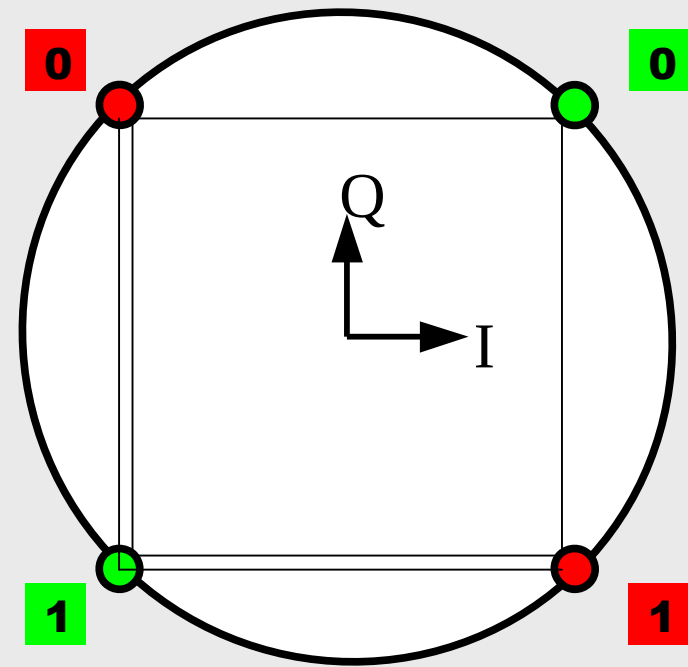
- Perspective
- What is needed for 3G
- Step 1 – Get the highest peak efficiency
- Step 2 – Extend the peak into the back-off region
- Step 3 – Minimise additional power to achieve spectral compliance
- Typical performance
- Summary

GSM Modulation

- BPSK modulation
- 1 bit per symbol
- Interleaved symbol sets
- Gaussian filtered

- Constant envelope
- 0 dB pk / average

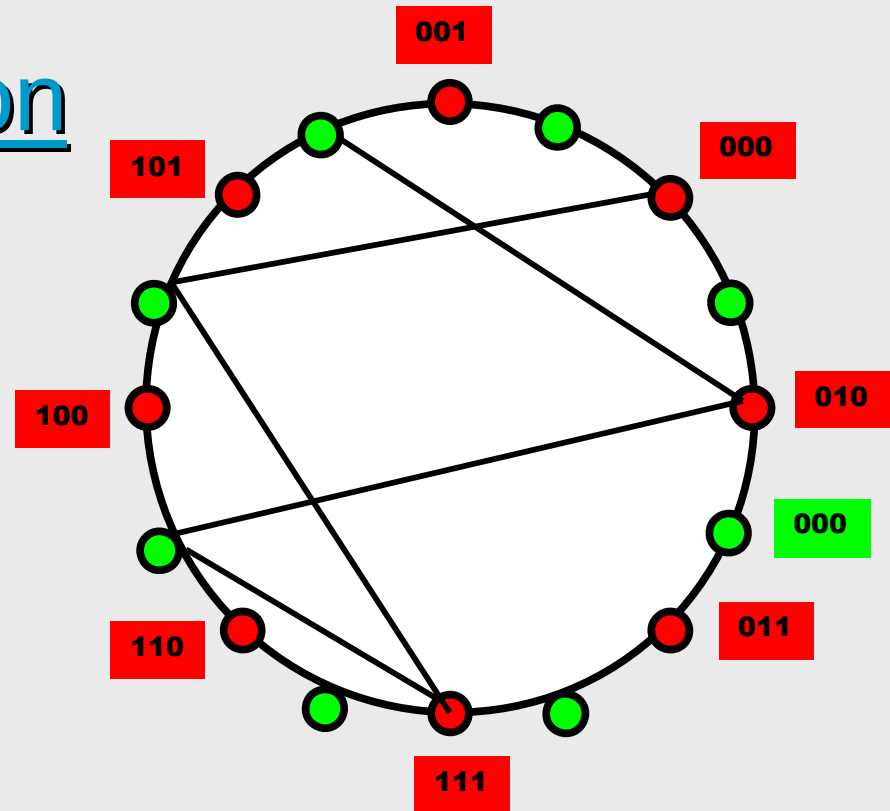
- 8 voice channels, time multiplexed on each RF carrier
- 200 kHz per RF channel
- 7:1 channel re-use



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EDGE Modulation

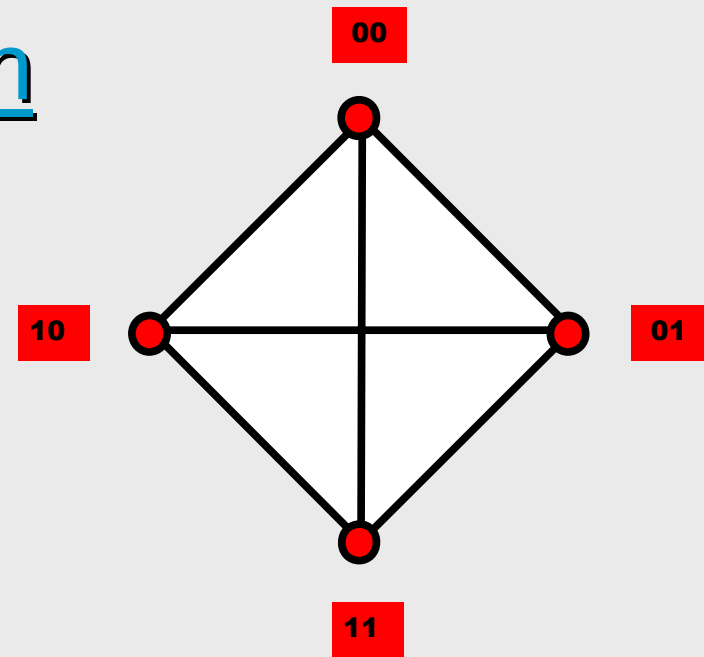
- 8 PSK modulation
 - 3 bits per symbol
 - Interleaved symbol sets
 - Gaussian filtered
-
- 3 dB pk / Average
-
- 8 signal channels time multiplexed on each RF carrier
 - 200 kHz RF channels
 - 7:1 channel re-use



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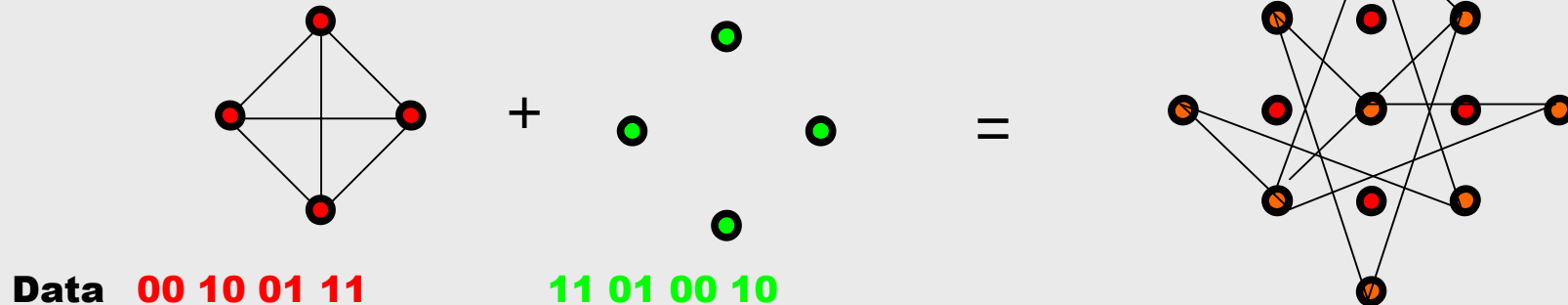
W-CDMA Modulation

- QPSK Modulation
- 2 bits per symbol
- Symbols are “coded”
 - multiplied by orthogonal codes
 - Added together
- Root raise cosine filtered
- >10dB pk/Average
- Many signal channels per 3.84 MHz carrier
 - e.g. 64 voice signals typical
- 1:1 frequency re-use



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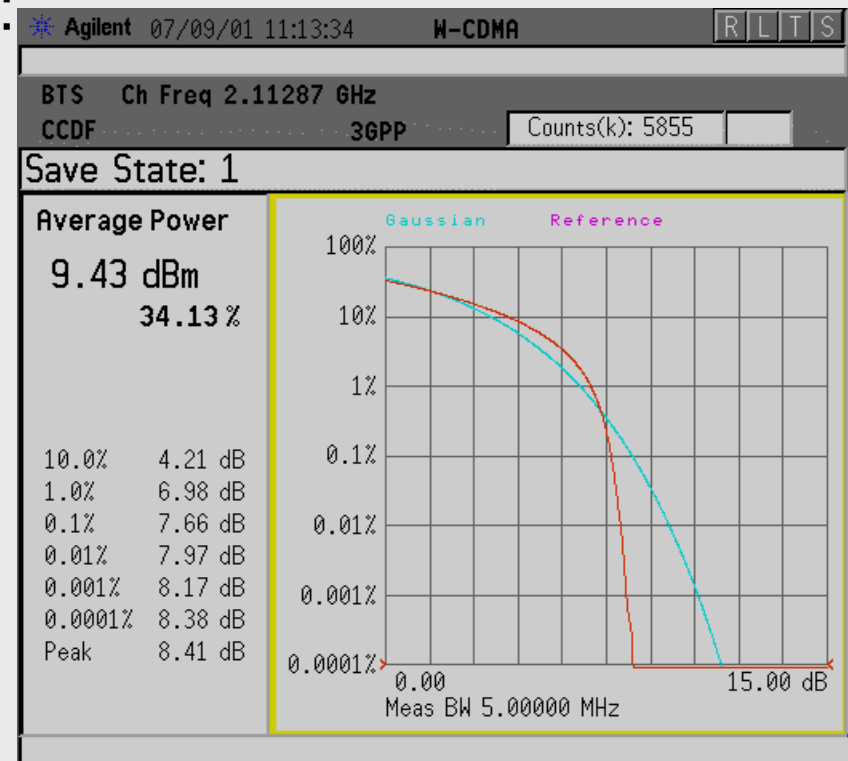
W-CDMA Coding



- Each QPSK signal is multiplied by one of a set of orthogonal codes
 - The “scrambled” signal is spread out in frequency
- All scrambled signal are added together, and transmitted
- The resulting signal has 3.84MHz BW and high peak to mean
 - Peaks are rare because they only occur when all the scrambled signals coincide with the same symbol

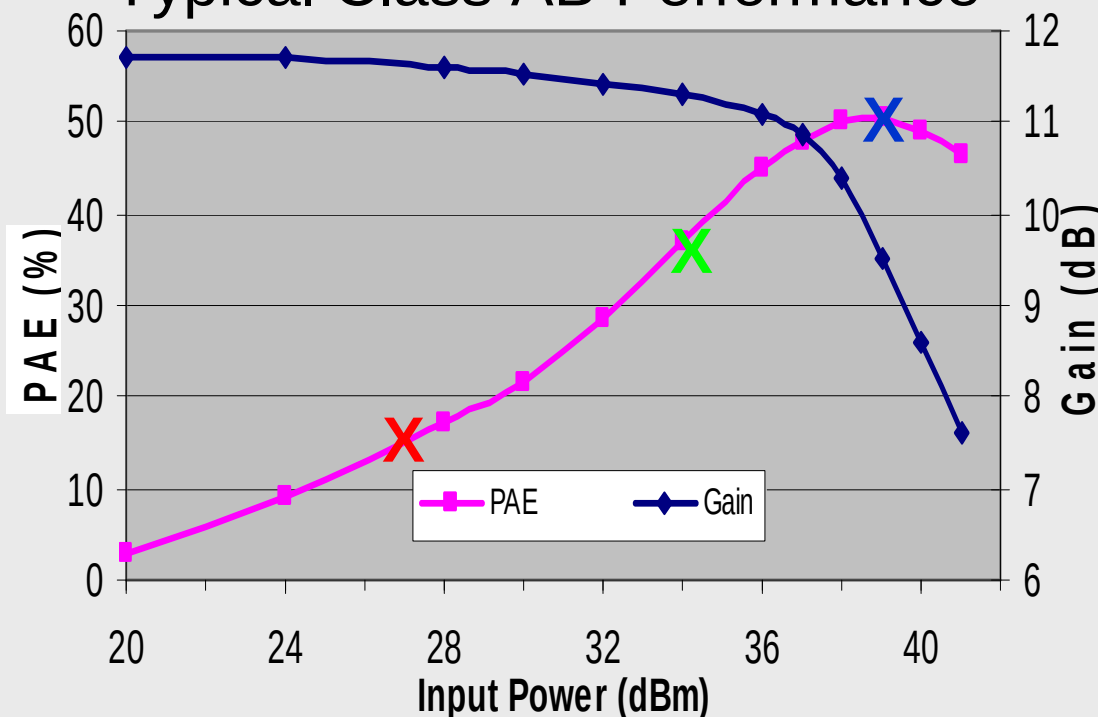
CCDF

- Cumulative Complementary Distribution Function
- The CCDF of a signal at x dB is:
 - The probability of the instantaneous signal level being x dB over the average
- For W-CDMA
 - CCDF at 0.01% is ~11.5 dB
- CCDF at 0.01% level
 - Also called PAR, peak to average ratio
- The dB level of 0.01% CCDF
 - Also called clipping factor



What does all this mean for the PA?

Typical Class AB Performance



- Independent of PA technology
- Max efficiency only at peak power
- For a given signal, the average efficiency is essentially the efficiency at the average power level

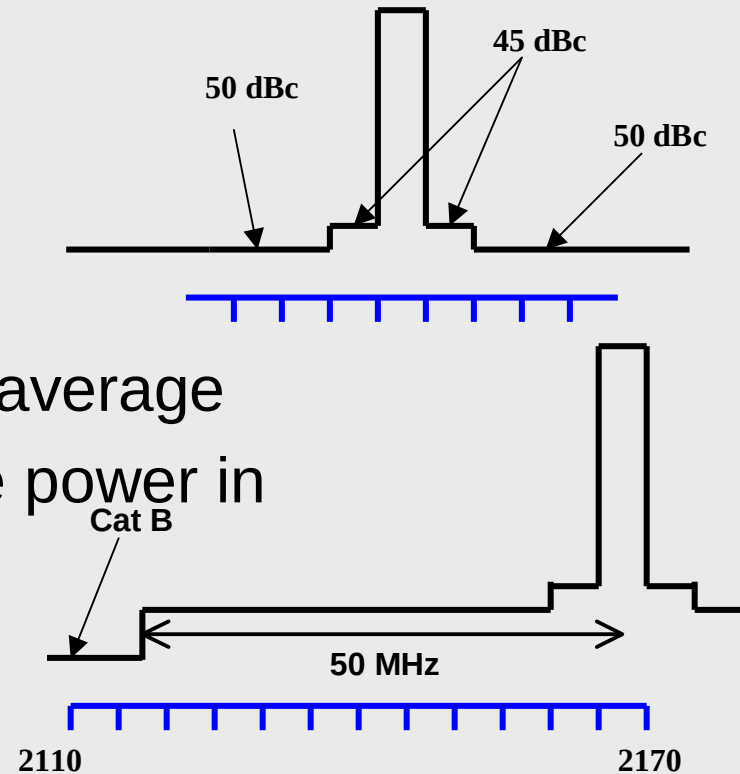
Typical values, excluding further back-off required for IM compliance

W-CDMA ~15% **EDGE** ~35% **GSM** ~50%



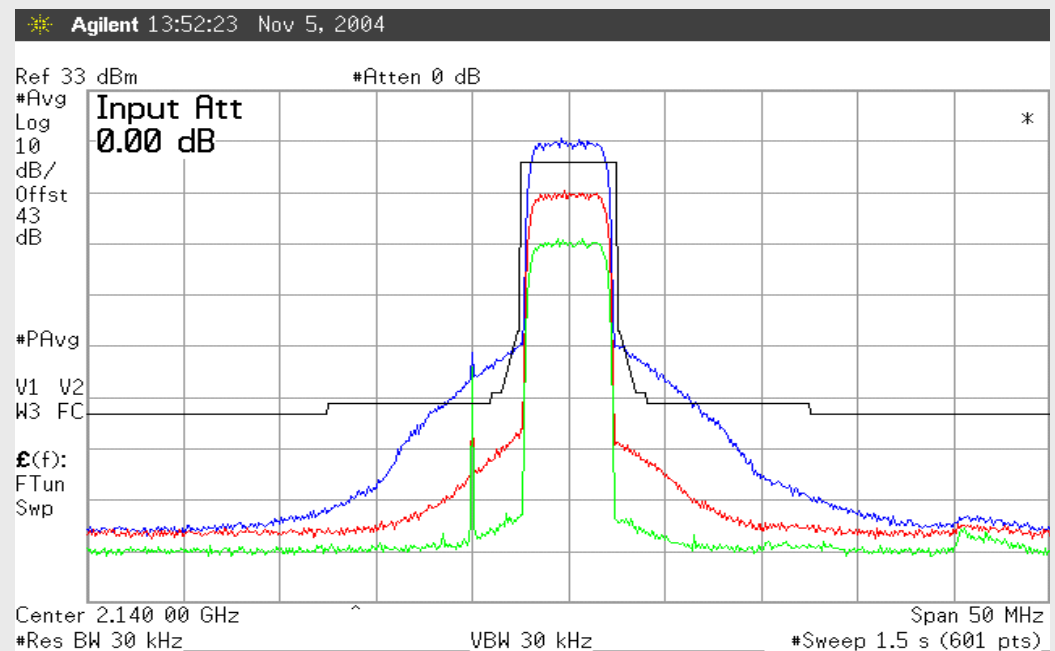
3G Requirements

- Signal BW 3.84 MHz
- Signal has ~11 dB peak to average
- Allowable IM is the average power in adjacent channels
 - -45 dBc +/- 5MHz
 - -50 dBc +/- 10Hz
- 2100, 2180MHz - Category B
 - -25 dBm, 50 MHz from carrier centre
 - *Problematic for carriers at the band edge as this is within the Tx filter passband*



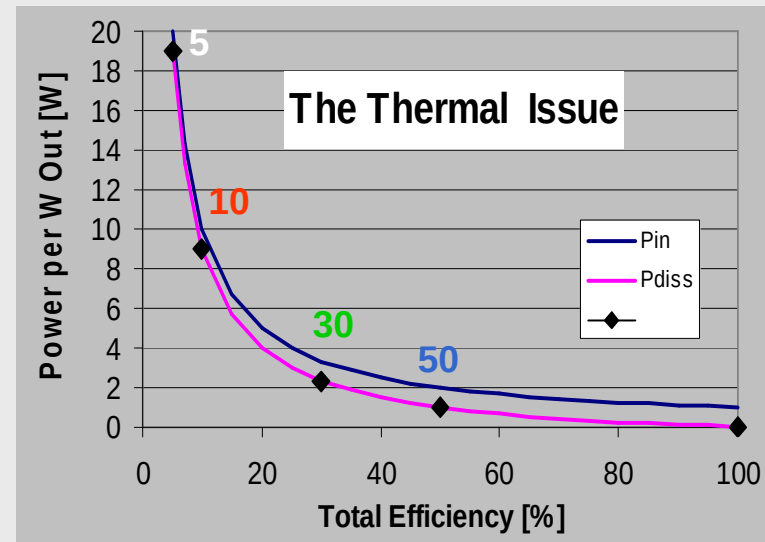
Typical Class AB Performance

- Single carrier W-CDMA signal
 - Spectral emissions mask superimposed
- Blue max power at 22% drain efficiency
 - Not linear enough
- Red and green show 10 and 20dBs backed-off
 - <10% efficiency

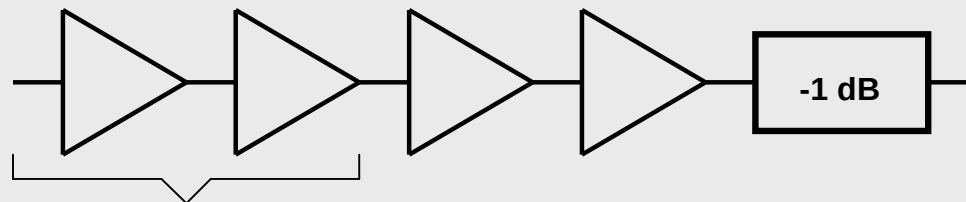


PA Efficiency - What is the Issue?

- Low Efficiency has a catastrophic affect on thermal dissipation
 - At 10% a 20W amplifier dissipates 180W as heat!
 - This is quite a shock when compared to GSM
- Operators consider efficiency as total DC in to RF out
 - Including any RF losses to the BTS antenna port
 - Including driver stages
 - *The exciter typically generates +5dBm*
 - And any DC the DC converter



Where Does all the Power Go?



RF = 20W

Total for amps 29.4%

Linearization (12W)

Linearization cuts that to 25%

- For a 20W PA with modest output losses
 - Actually need 25.2 W of RF from the amplifier
 - At 45% average drain efficiency, that's 56W of DC
- And ~2.5 Watts drive
 - At good linearity (25%) another 10W
- Also need 2 more stages of gain
 - Another 2W or so
- Then some form of linearization (12W)

Key Observations

- W-CDMA signal has a high peak to average
 - Average amplifier drain efficiency is essentially the CW efficiency at the signal average level
- To get to the overall amplifier efficiency
 - Further power needed for all the drivers and
 - Further back-off or additional linearization needed to meet spectral requirements
- This is a big shock to GSM operators, used to >50% overall amplifier efficiency
- Even to get to 25% overall, need 45% drain efficiency
 - Impossible with a simple class AB approach

Three Steps to Higher Efficiency

- Step 1, Get the highest peak efficiency
 - Higher peak efficiencies imply higher efficiencies at all levels of back-off
- Step 2, Improve efficiency with back-off
 - i) Pull up the average
 - ii) Extend the peak from the top
- Step 3, Spectral compliance with minimal extra power
 - Allows operation to the maximum level of efficiency

Get the Highest Peak Efficiency

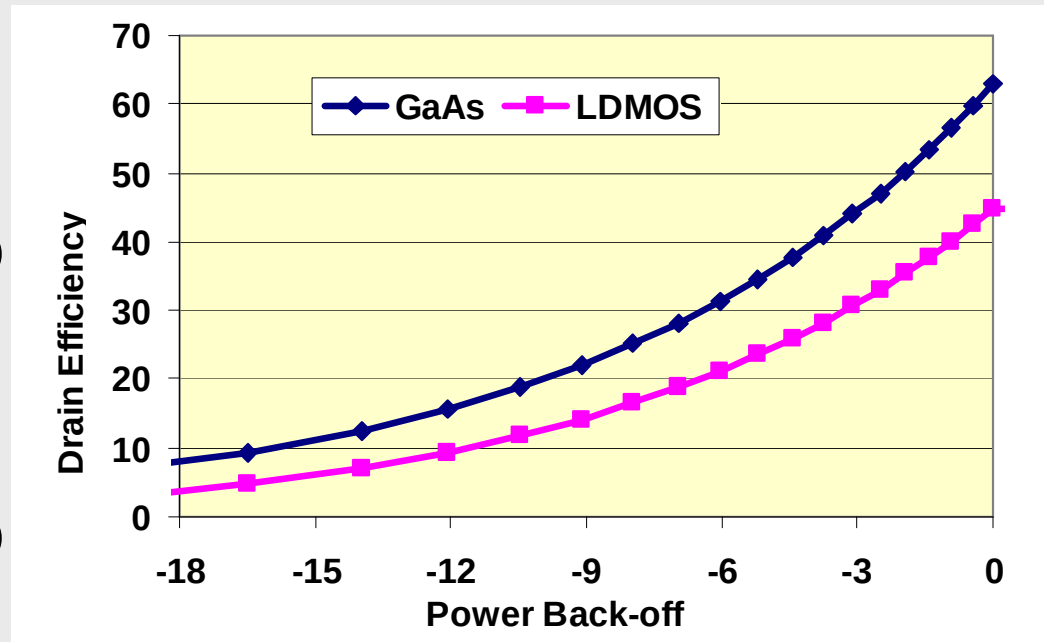
■ Comparison of peak drain efficiencies

■ GaAs

- Typically >60%
- ~20% at 10dB BO

■ LDMOS

- Typically 45%
- ~12% at 10 dB BO



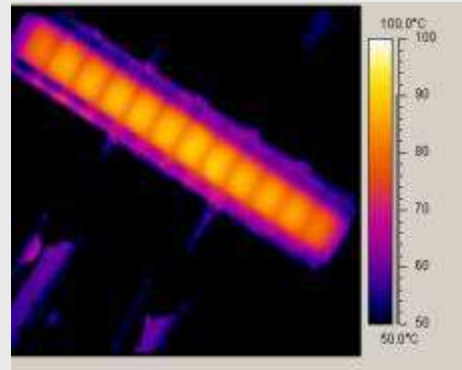
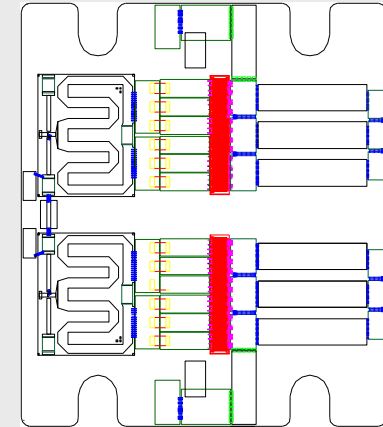
■ Every little helps, but this on its own is far short of the >45% target



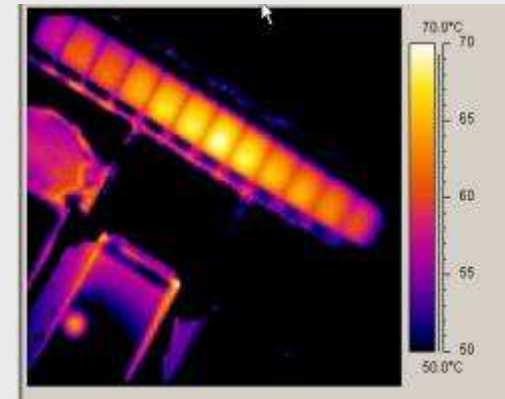
CW Performance for GaAs Amplifier

- 2 x 144 mm FETs
- Best power
 - Pout 150W, with 56% efficiency
- Best linearity
 - Pout 130W, with 63% efficiency

- Thermal Images



Bottom Device 5A DC only



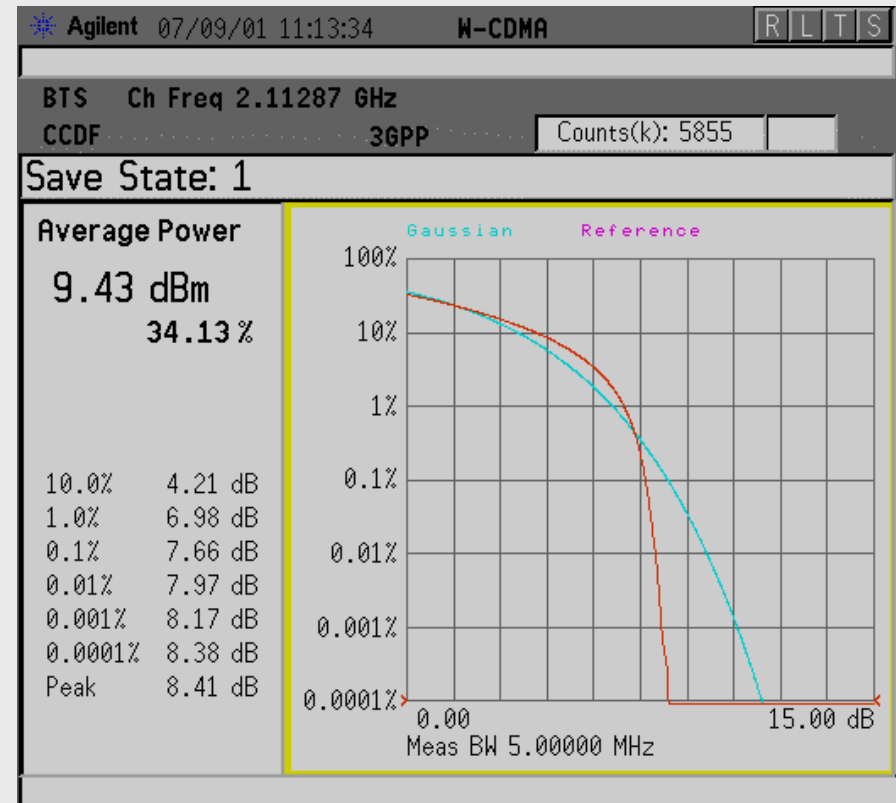
Bottom Device 150W RF out

Improve Efficiency with Back-off

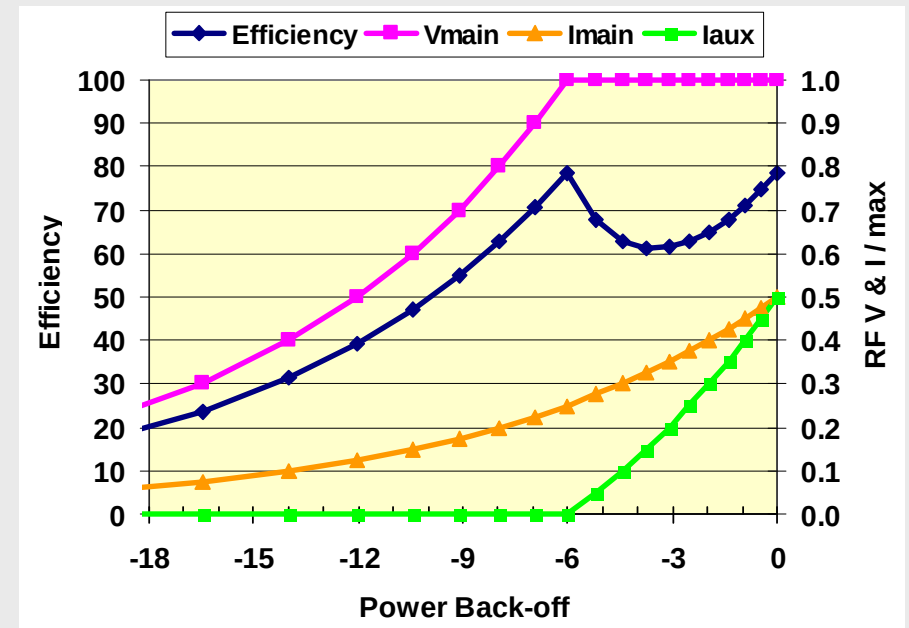
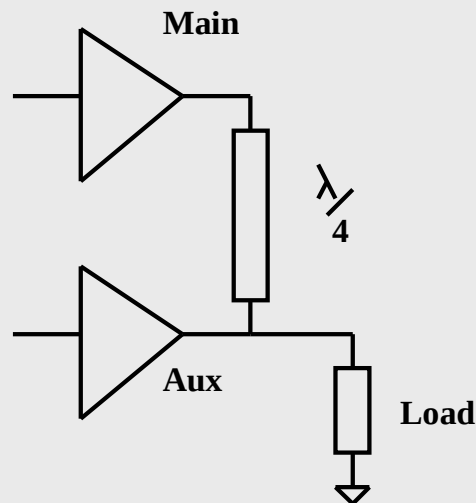
- Digitally eliminating the biggest peaks
 - Signal clipping, pre-conditioning, reduced PAR
 - Out of band distortion can be filtered out in the baseband
 - In band, contributes to residual EVM, but tolerable to a reasonable level
- Circuit techniques to extend peak efficiency down from peak power
 - Doherty amplifier
 - Out-phasing, Chirex, Envelope elimination and others

Signal Clipping

- Maximum practical clipping is ~8dB above average.
 - Heavier clipping degrades EVM too much
 - Many operators use 8.5dB
- This allows up to 3dB more peak power
 - At least 5% better average efficiency
- Watch out for results quoted with excessive clipping



Classic Doherty Amplifier

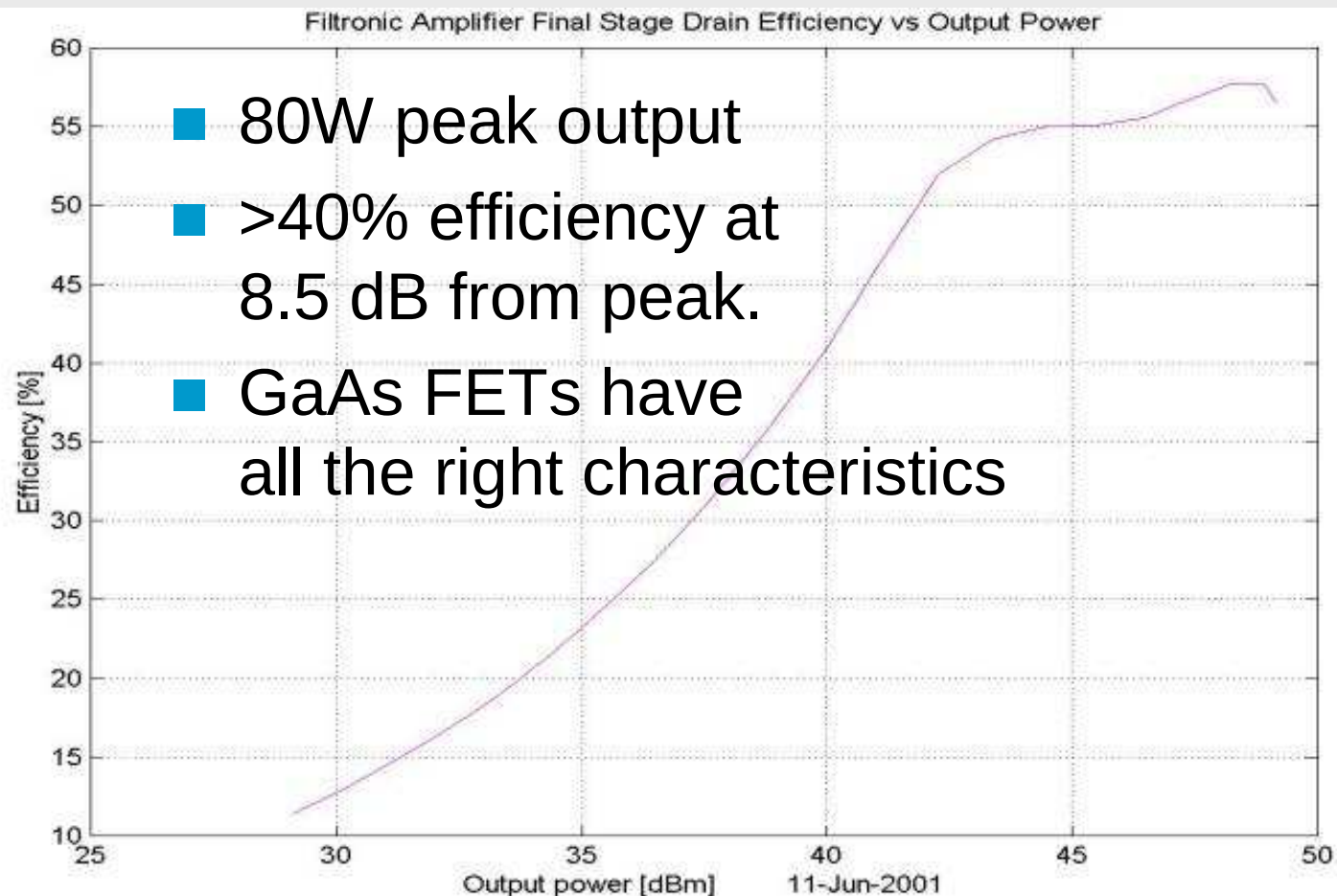


- Main and Aux are equal sized amplifiers
- Extends the peak efficiency to 6 dB back off (BO)
- Theoretically >50% at 8.5 dB BO

Doherty Amp. Key Points

- Up to 6dB back-off
 - Aux amp. is off, must provide an open circuit
 - Main amp works into twice the optimum load
 - Normal class A/B efficiency characteristic
 - The Aux amp must be very low loss when pinched off
- Above 6dB
 - Aux contributes to output power
 - Main sees the load reduce to the optimum value
 - Main amp. must produce peak efficiency into a variable load

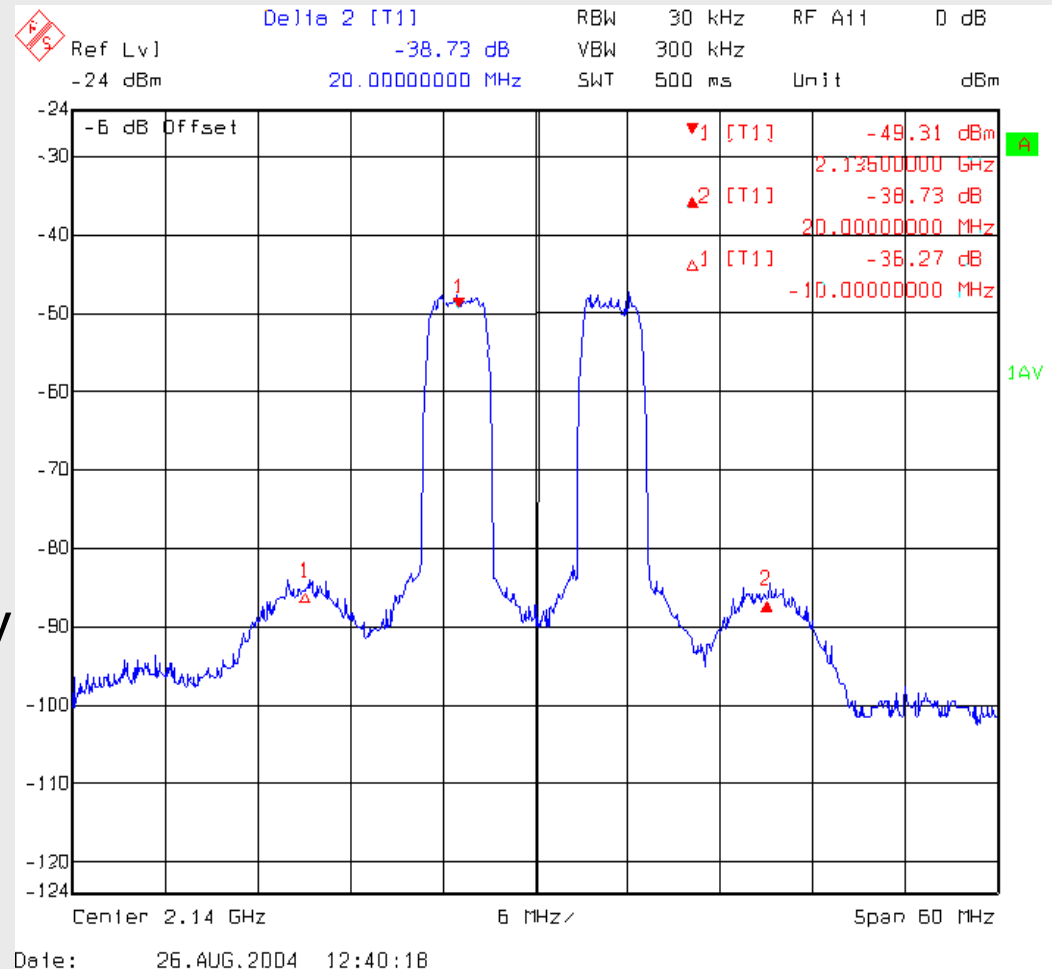
GaAs Doherty CW Efficiency



- 80W peak output
- >40% efficiency at 8.5 dB from peak.
- GaAs FETs have all the right characteristics

GaAs Doherty W-CDMA Performance

- Un-corrected linearity
- W-CDMA at 8.5 dB PAR
- 20 W average with drain efficiency 25%
- Raw linearity 36dBc
- Some signs of asymmetry
 - Aka memory effects



Spectral Compliance with Minimal Extra Power

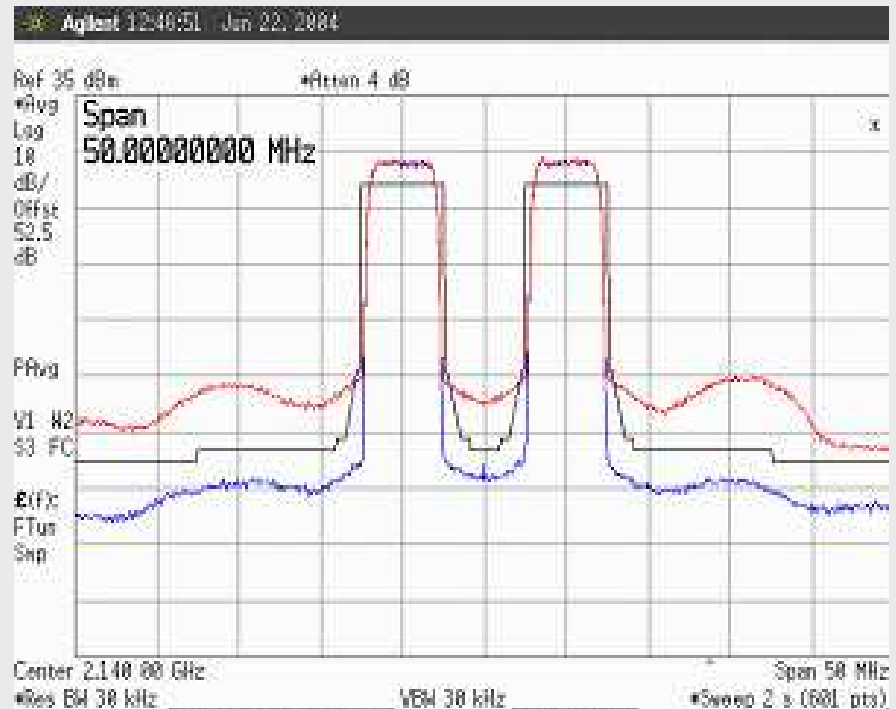
- Various techniques possible
- Simple power back-off
 - Loss of efficiency generally excessive
- Feed-forward linearisation
 - Well proven technique, still popular
 - Requires an error amp which adds cost and power
 - Also couplers which add cost and degrade output power
- Digital Pre-distortion
 - Yet to be proven
 - Requires access to digital base-band information

Digital Pre-Distortion (DPD)

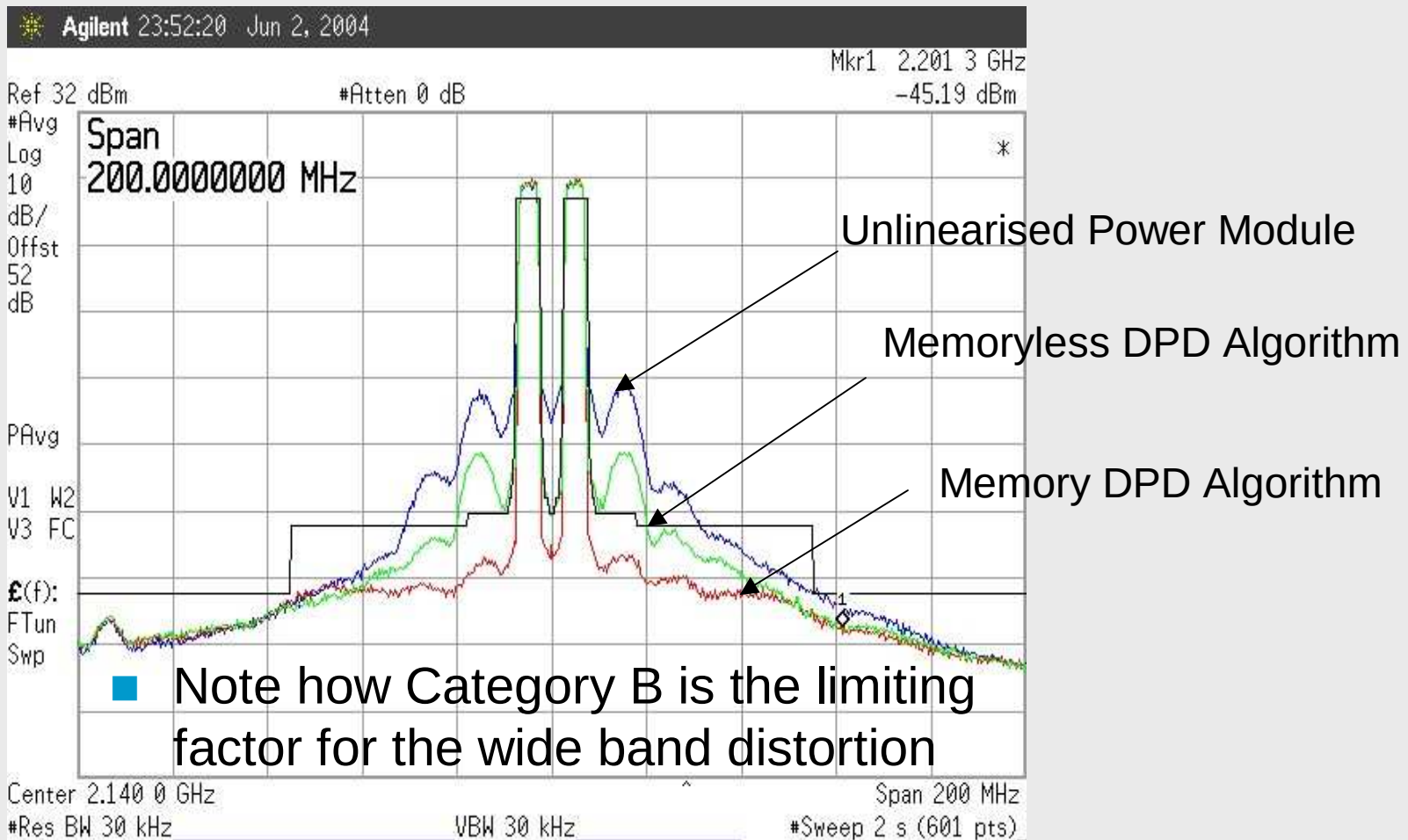
- This is regarded as ‘the holy grail’
 - Fortunately recent advances show it is not so elusive
- Completely consistent with the drive towards digital radio solutions
 - Digital baseband processing is now quite common
 - Adding DPD is a natural fit, with low incremental overhead
- Minimises RF Hardware
 - DPD basically eliminates RF complexity that would otherwise reduce the efficiency of any amplifier
 - *Output losses eg couplers phasing etc, also error amps.*
- Other key benefit is that it saves on material cost.
 - Digital hardware is cheap, low power and going down.

DPD Results

- Before and after DPD
- 50W av at 33%
- Starting from a good uncorrected linearity
- Can improve by 15 – 18 dB
- Also note the improvement in asymmetry
 - DPD model includes memory effects
- Contrary to popular belief
 - GaAs FETs have good inherent linearity
 - AND can be pre-distorted as well as other devices



2 x 20W WCDMA Cat. B

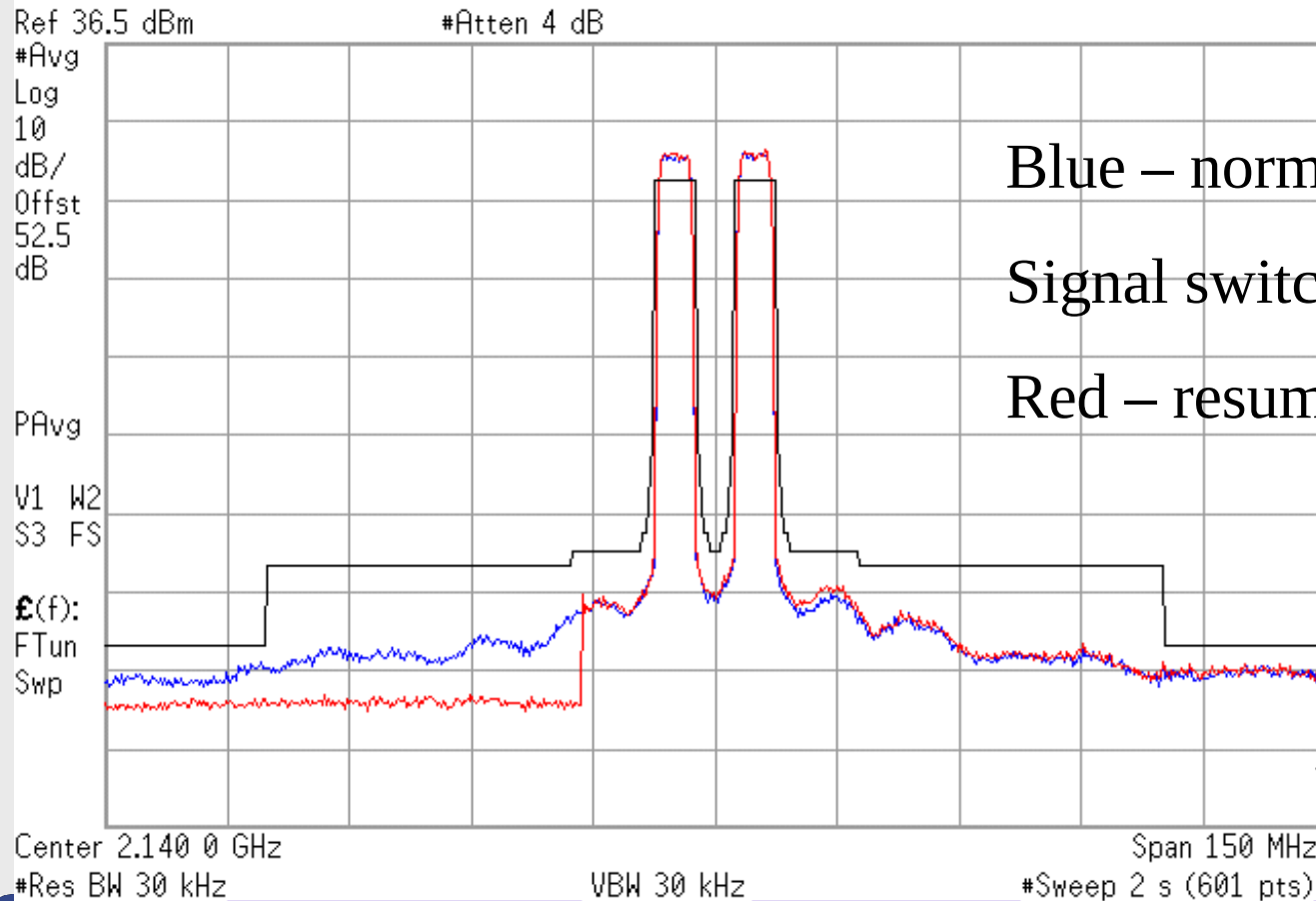


DPD Caution

- Operational issues
 - A distortion receiver is needed to observe the corrected output.
 - *Distortion bandwidth is wide by nature*
 - Allows update of coefficients for thermal drift
- Dynamic signal tracking
 - The transmitter average power also varies, especially for HSDPA
 - *Coefficient updates cannot track these quite rapid changes*
 - *Long memory effects must be miniscule*
- Convergence
 - Numerical noise
 - *The error is the difference between big signals*
 - Lower average power is not always lower ACPR
- Category B spectral requirement
 - This is a limit on far out distortion that is virtually un-correctable.

DPD GaAs Amplifier under Dynamic Signal Conditions

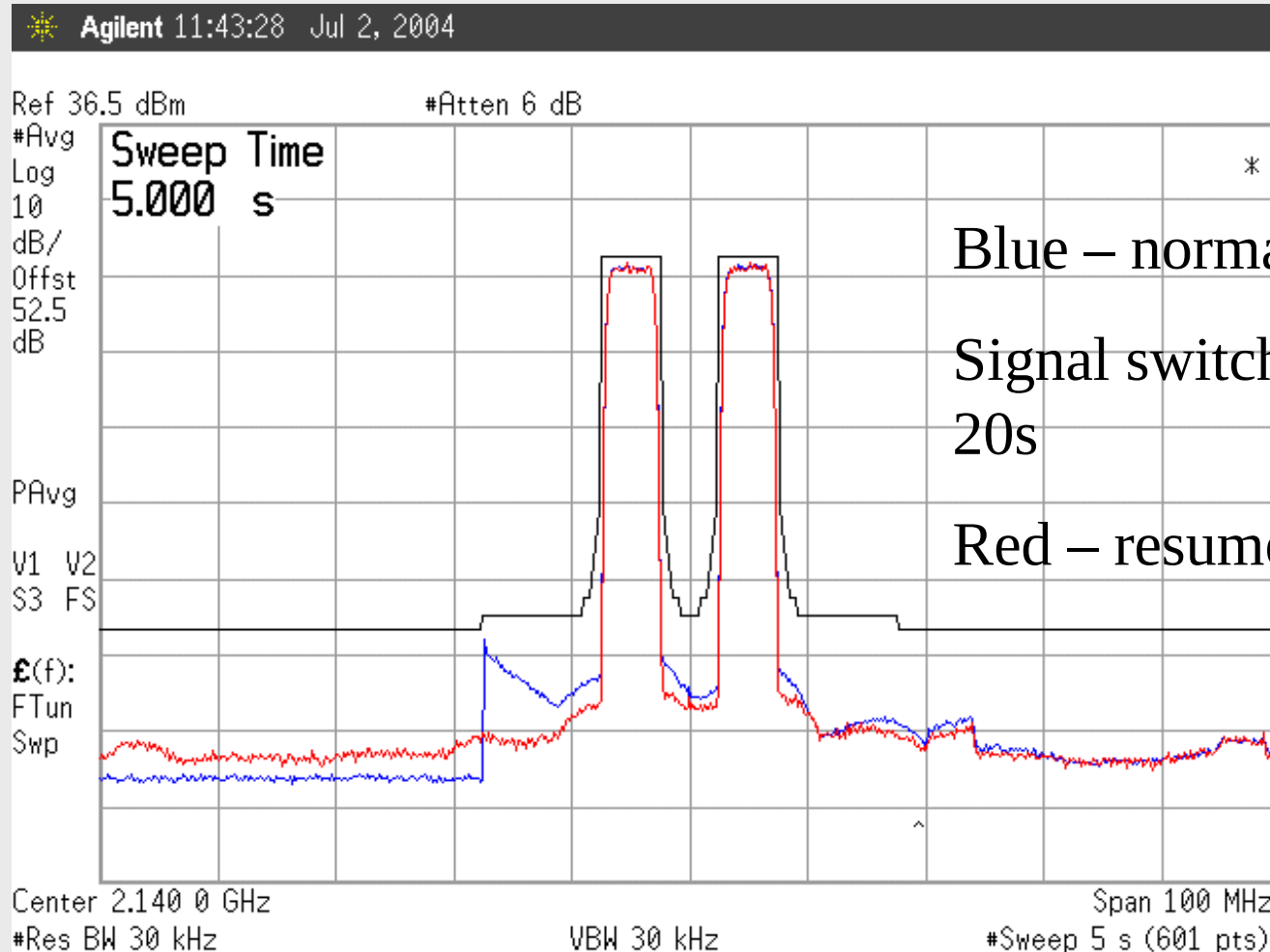
Agilent 17:09:30 Jul 1, 2004



Blue – normal DPD setup
Signal switched off for 20s
Red – resumed sweep



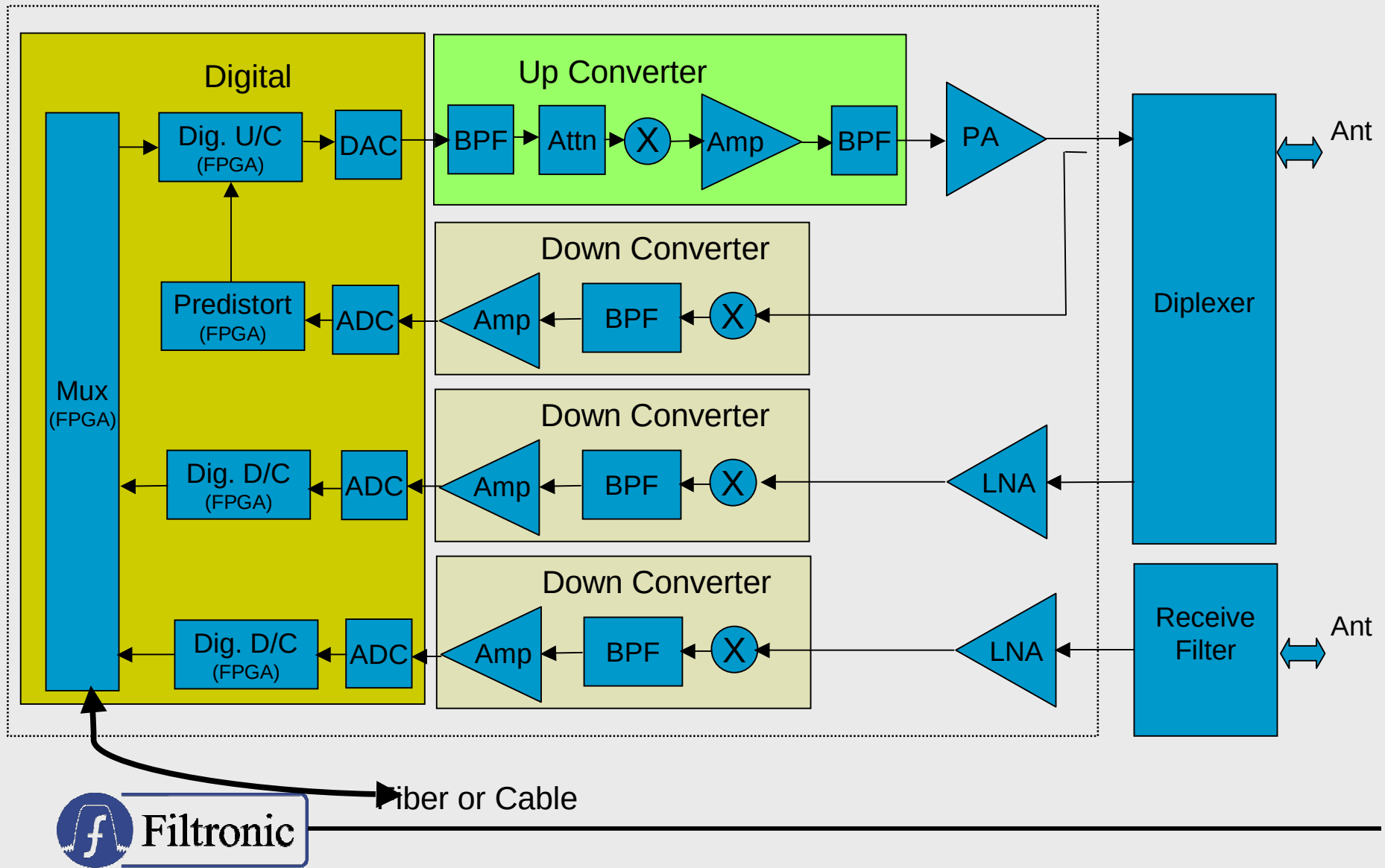
DPD LDMOS Amplifier under Dynamic Signal Conditions



W-CDMA Transceiver/PA

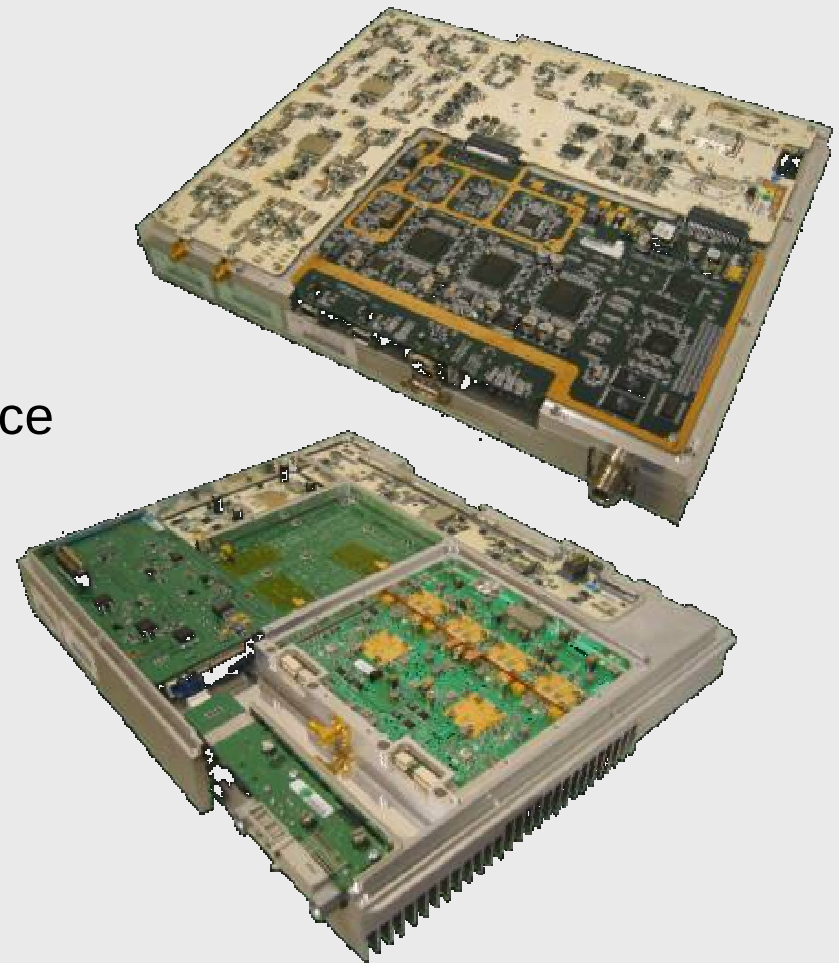
- Entire active RF section of a BTS
 - Transmitter chain, D→A converters, upconverter and PA
 - Receiver chain, main and diversity channels, downconverters and A→D converters
 - CPRI Digital baseband interface
- Includes a distortion receiver, remote control via TCP/IP and operation from 48V DC
- 30W W-CDMA output
 - Complete unit efficiency >14%
 - Extracted feed forward PA efficiency >19%

RF Head Architecture



CPRI Compliant Integrated Transceiver/PA

- 30 Watt high efficiency FF
- CPRI I/Q base band input
- DSP implemented in FPGA
- Embedded LINUX operating system
- Ethernet via HDLC control interface
- 15 MHz Operating BW within 60 MHz UMTS band
 - 1 TX channel (1 or 2 carriers)
 - 2 x RX channels w. diversity
 - Expandable to 3TX, 3RX



Summary

- Efficiency is total DC power input for a certain RF power out
- Device peak efficiency is important
 - GaAs devices have consistently the highest efficiency
- Maintain efficiency with power back off
 - Doherty amplifier gives high peak efficiency at 8.5 back-off
 - GaAs devices work well in the Doherty configuration
- To use the peak efficiency some linearization is needed
 - Digital pre-distortion offers the lowest power penalty
 - GaAs device can be pre-distorted to the required level
- Demonstrated a ~20% efficient W-CDMA amplifier using FF and >30% is practical using DPD